MONASH University

Basketball Analysis - UBL women's basketball, 2025

ETC5543 - Business analytics creative activity

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GitHub repository

github.com/mey63631/Basketball-Project

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1 Abstract (Summary)

Coaches substitute players constantly, but individual statistics such as points and plus-minus cannot show which combinations work well together, because a player's results also depend on her teammates, her opponents and her playing time. This project examines player-pair performance and rotation structure in La Trobe University's (LTU) women's team in the 2025 University Basketball League (UBL). Event-level play-by-play data were used to reconstruct five-player lineups and same-lineup intervals ("stints") for 8 of 10 games.

A duration-weighted ridge regression (RAPM-style) estimated net points per minute from the LTU players on court and the opponent team. A pair's connection score was the duration-weighted average gap between actual and predicted performance in the stints the pair shared, reported for pairs with at least 15 minutes together across at least two games. Of 73 eligible pairs, 35 were stable positive and 28 stable negative under leave-one-game-out refits, and Player 7 appeared in 3 of the five highest-scoring pairs. Adding pair scores did not clearly improve prediction of held-out games (duration-weighted RMSE 2.303 versus 2.300), so the scores are best used to prioritise video review, not to change rotations. Interpretation is limited by the small sample, reconstructed lineups and the use of minutes rather than possessions.

2 Background and Motivation

Coaches must decide which players to use together and when to change a lineup. Individual statistics describe aspects of player performance, but they do not establish how particular combinations perform with each other. A player's plus-minus also reflects the teammates, opponents and game situations encountered during that particular player's minutes on court (Rosenbaum 2004).

Adjusted plus-minus addresses part of this problem by regressing scoring margin on indicators for the players on court (Rosenbaum 2004). Because players repeatedly appear together, the indicators are strongly correlated and estimates can be unstable. Ridge regression penalises large coefficients to stabilise estimation (Hoerl and Kennard 1970), and Sill (2010) motivated regularisation and out-of-sample testing for adjusted plus-minus. Network methods offer a complementary view: players or teams are nodes, and centrality measures describe their position in the network (Wasserman and Faust 1994). Eigenvector-based methods have also been used to rank teams from game results (Keener 1993).

Most adjusted plus-minus work targets individual ratings. Pairs are harder, because the number of parameters grows quickly while the number of games stays small. For LTU women's basketball, an analysis of shared playing time offers a way to organise evidence about combinations and rotation patterns. Positive performance during a pair's shared minutes may indicate a combination worth investigating, but it can also reflect with the other three teammates, favourable match-ups or short-term scoring variation. Pair rankings therefore need exposure information and validation before they can inform coaching decisions.

This project combines an adjusted performance summary with a network description of player deployment. It separates three questions: which combinations were associated with above-prediction performance, how players were connected through shared minutes, and whether the estimated pair scores improved predictions in other games.

3 Objectives and Contribution

The primary question is: during which LTU pairings did the team perform better or worse than predicted by an additive model of its on-court players and opponent team?

The analysis has four objectives:

1. Reconstruct on-court lineups from event-level data and document the reliability of that reconstruction.
2. Calculate pair connection scores relative to an additive lineup model, reporting shared minutes and games alongside each score.
3. Describe rotation structure through a network weighted by shared playing time, and examine exposure across game situations.
4. Assess sensitivity to individual games and evaluate whether a pair-score adjustment improves held-out prediction.

The practical contribution is a workflow for turning play-by-play records into evidence that can guide coaching review. The analytical contribution is the integration of lineup reconstruction, regularised modelling, network summaries and game-level validation within a small single-team dataset. The project does not identify causal player chemistry or an optimal rotation.

Project contributions

- **Data collection:** a script that retrieves fixtures and play-by-play events from the competition feed, adapts to a change in the feed's response format, and saves extracted input tables for subsequent analysis.
- **Data processing:** a lineup and stint reconstruction algorithm with built-in checks, including a game-coverage screen and a comparison of reconstructed scores with the feed's scoring events.
- **Modelling and validation:** a regularised lineup model, pair connection scores, leave-one-game-out stability and a held-out prediction test.
- **Networks and visualisation:** shared-minute networks with centrality and concentration measures, and anonymised figures for reporting.
- **Reproducibility:** analysis scripts that run in a fixed order and save the outputs this report reads (Appendix A).

4 Data and Preparation

Source and scope

The data concern LTU women's games in the 2025 UBL. Fixture and play-by-play records were supplied by the Sportradar feed used for the competition. Fields include fixture identifiers, period, game clock, team, player identifier, event type, event subtype, substitution direction and scoring outcome. Persistent player identifiers are used for processing. To protect privacy, players are reported as Player 1 to Player 16 throughout.

The saved play-by-play input contains 10 LTU games and 5,936 events. Of these games, 8 were retained after excluding games with reconstructed LTU playing-time coverage below 50% of an assumed 40-minute game. Among the retained games, 6 had complete reconstructed coverage and matching scoring totals. The other 2 games had coverage ranging from 79.0% to 84.5%. Retention therefore does not establish complete coverage or correct lineup identities. Table 1 summarises coverage and scoring checks for the retained games.

Two of the ten games were excluded because lineup tracking failed: in each, a starter had no recorded action before her team's first substitution, so the opening lineup was incomplete and errors accumulated through the game (reconstructed coverage about 13% of regulation time, against at least 79% for retained games). League-wide results for both competitions come from the same feed.

Table 1: *Reconstruction checks for retained LTU games. Coverage is reconstructed playing time as a percentage of an assumed 40-minute game. Point totals are shown as LTU–opponent.*

Game	Coverage (%)	Reconstructed points	Feed points	Check
Game 1 vs VU	100.0	84–62	84–62	Match
Game 2 vs AU Waatu	84.5	79–60	89–72	Investigate
Game 3 vs MELB	100.0	75–57	75–57	Match
Game 4 vs UON	79.0	55–47	68–64	Investigate
Game 5 vs CU	100.0	74–72	74–72	Match
Game 6 vs UWA Maali	100.0	82–60	82–60	Match
Game 7 vs UniSC Mari	100.0	93–83	93–83	Match
Game 8 vs MELB	100.0	62–99	62–99	Match

Lineup reconstruction

Within each game, events are ordered by period and descending game clock, preserving source order for tied timestamps. Opening lineups are inferred from distinct players with

recorded actions before the team's first substitution. Where five starters cannot be identified, lineup tracking may fail or be incomplete.

Substitutions update the inferred on-court roster. Consecutive substitution events at the same timestamp are processed together, and stints are closed at substitution times and period boundaries. A stint is a continuous interval during which the inferred lineup is unchanged. Points scored and conceded (made two-point and three-point shots, and made free throws) are accumulated within each stint.

The final lineup of a period is carried into the next period. The implementation checks for activity inconsistent with that lineup but flags discrepancies rather than correcting them. This is an assumption of reconstruction procedure: players can change between periods without an in-period substitution record. The reconstruction procedure flagged no quarter-break lineup discrepancies in the saved-input analysis. This does not independently verify lineup identities.

Stints shorter than 15 seconds are excluded from regression and connection scoring to reduce extreme rate estimates. The rotation-network calculations use a separate sample of all positive-duration stints. The two samples therefore differ in total minutes and pair counts.

Data provenance and quality

The data were retrieved from the Sportradar feed embedded in the UniSport UBL website. All results in this report use the saved input snapshot `data/raw/LTU_women_inputs_2025.rds`, containing 5,936 play-by-play events across 10 LTU games. The snapshot records its export time as 2026-11-01 14:03:52.80575; this is not necessarily the original collection date. Rendering the report does not download new data.

The extraction function was adapted to handle different API response structures encountered during the project. Because the live feed may change, the saved inputs are used to keep the analysis consistent across reruns.

Reconstruction checks compare accumulated stint scores with totals calculated from scoring events in the same feed. Scoring totals matched in 6 of 8 retained games (Table 1). The checks recorded 0 negative-duration stints. 2 retained games had incomplete reconstructed coverage and differences in scoring totals, which may affect player and pair estimates.

These checks assess internal consistency rather than accuracy against an independent source. Complete playing-time coverage and matching scores do not verify every lineup: an inferred lineup can contain five players while identifying the wrong five. No independent validation against official scores or video is established by these checks.

5 Methodology

Response and additive lineup model

For stint s, net points per minute is

$$y_s = \frac{P_{s,\mathrm{LTU}} - P_{s,\mathrm{opp}}}{d_s/60}, \tag{1}$$

where P denotes points scored in the stint and d_s is stint duration in seconds. The additive model is

$$y_s = \beta_0 + \sum_p \beta_p x_{sp} + \sum_o \delta_o z_{so} + \varepsilon_s, \tag{2}$$

where $x_{sp} = 1$ if LTU player p is on court and $z_{so} = 1$ if the opponent is team o. The model uses duration weights, so longer stints contribute more to fitting.

Ridge regression is used because the player indicators are strongly related: players repeatedly appear together, and each valid lineup contains five players. The model is fitted with the `glmnet` package (Friedman, Hastie, and Tibshirani 2010) using $\alpha = 0$, which minimises (up to the scaling used by `glmnet`)

$$\sum_s d_s \Big(y_s - \beta_0 - \sum_p \beta_p x_{sp} - \sum_o \delta_o z_{so} \Big)^2 + \lambda \Big(\sum_p \beta_p^2 + \sum_o \delta_o^2 \Big). \tag{3}$$

Entire games are assigned to cross-validation folds, and the penalty λ is the largest value whose cross-validation error is within one standard error of the minimum (`lambda.1se`).

This is a single-team RAPM-style model, not a full model of players on both teams. The opponent indicators identify the opposing team but not which opposing players were on court. The code penalises the opponent coefficients along with the player coefficients, so they are regularised opponent-team controls rather than unpenalised fixed effects.

Because the player indicators sum to five and the opponent indicators sum to one in every stint, the unpenalised design is redundant with the intercept. Ridge regularisation still yields a fitted solution, but individual coefficients depend on the penalty and coding. They should not be read as isolated causal contributions of individual players.

Pair connection scores

The fitted model gives a residual $y_s - \hat{y}_s$ for each stint. For players i and j, the connection score is

$$C_{ij} = \frac{\sum_{s:\, i,j \in L_s} d_s \, (y_s - \hat{y}_s)}{\sum_{s:\, i,j \in L_s} d_s}, \tag{4}$$

where L_s is the inferred LTU lineup in stint s. A positive score means LTU performed above the additive lineup prediction during that pair's shared stints. Its unit is net points per minute above prediction.

The prediction behind each residual includes the intercept, all five LTU player indicators and the opponent indicator. The score is therefore not raw pair performance minus two player ratings. It is also not a separately estimated pair-interaction coefficient: each five-player stint contributes the same residual to ten overlapping pairs. The residuals are in-sample.

Pairs are reported when they share at least 15 model-eligible minutes across at least two games. These thresholds reduce reliance on brief appearances but do not guarantee precise estimates, so each score is reported with minutes, games and a stability summary.

Leave-one-game-out sign stability

Each retained game is removed in turn, the model is refitted, and eligible pair scores are recalculated. A pair is classified as stable positive if its score is positive in at least 80% of available refits, and stable negative if that proportion is at most 20%. Other pairs are classified as uncertain. Pairs with fewer than half as many estimates as retained games are labelled as having insufficient stability evidence; with eight games, at least four valid estimates are needed.

These labels describe sensitivity to omitting a game. They are not confidence intervals or significance tests, and they are not evidence of future predictive performance. Each refit reuses most of the same games, so the labels are far from independent validation.

Rotation networks and game situations

A separate undirected usage network has players as nodes and shared playing time as edges. A pair enters the network after at least one shared minute in at least one game, unlike the stricter connection-score threshold. Weighted strength is the sum of a player's edge weights. Weighted betweenness is calculated using inverse shared minutes as network distances, so frequently used pairings are treated as shorter connections (Freeman 1978).

High betweenness indicates a position on shortest paths in this shared-minute network. It does not measure ball movement, leadership, passing skill or causal importance. Weighted strength is closely tied to exposure, since each minute played adds roughly four shared-player minutes.

Concentration is summarised with the Gini coefficient, where 0 means equal and 1 means fully concentrated. Freeman's betweenness centralisation is computed on the unweighted network, because its normalising constant is derived for unweighted graphs (Freeman 1978).

Game situations use the score margin at stint start: leading by at least six, within five points, or trailing by at least six. A clutch situation is a stint starting in the final five minutes of the fourth period with an absolute margin of at most five. A stint that crosses a threshold is not split.

Held-out prediction

In each outer evaluation fold, one game is held out. The baseline model, including penalty selection, is fitted on the other games, and pair scores are also computed from those games. For each test lineup, the ten pair scores are averaged (missing scores count as zero) and added to the baseline prediction. This is a fixed-coefficient heuristic, not a fitted interaction model.

Predictions are compared using duration-weighted root mean squared error,

$$\mathrm{WRMSE} = \sqrt{\frac{\sum_s d_s (y_s - \hat{y}_s)^2}{\sum_s d_s}}. \tag{5}$$

As an exploratory benchmark, the held-out pair feature is shuffled across pooled stints 2,000 times and the error improvement is recomputed. The reported p-value is the share of shuffled improvements at least as large as the observed one. Pooled shuffling ignores dependence within games, so this p-value is indicative rather than definitive.

Test players and opponents not seen in training receive no coefficient. Most opponents were met in only one retained game, so for those games the opponent term is unavailable and the baseline relies on the players and the intercept.

Software is listed in Appendix A.

6 Results

Pair performance and stability

Table 2 and Figure 1 show the highest and lowest connection scores. Of the 73 eligible pairs, 35 were stable positive, 28 stable negative, 10 uncertain and 0 had insufficient evidence. Scores ranged from -1.45 to 1.12 net points per minute relative to prediction. Player 7 appears in 3 of the five highest-scoring pairs. The model's individual ratings were small, ranging from -0.000 to 0.000, which is consistent with strong shrinkage.

The extremes are selected from many pairs, so they probably overstate any real effect. The stability labels also reuse the same games, so they show sensitivity to single games rather than evidence of repeatability.

Table 2: *The five highest and five lowest pair connection scores. Actual and predicted are net points per minute during shared stints; score is their difference. Refits is the number of leave-one-game-out refits in which the pair was eligible.*

Pair	Min	Games	Actual	Pred.	Score	Stability	Refits
Player 16 + Player 7	24.97	3	1.28	0.17	1.12	Stable +	7
Player 6 + Player 7	17.02	3	1.23	0.17	1.07	Stable +	5
Player 10 + Player 12	15.22	2	1.18	0.17	1.02	Stable +	6
Player 12 + Player 1	15.17	2	0.92	0.17	0.76	Stable +	6
Player 4 + Player 7	59.38	4	0.89	0.17	0.73	Stable +	8
Player 13 + Player 14	15.57	4	-1.28	0.17	-1.45	Stable -	4
Player 8 + Player 14	33.80	3	-0.98	0.17	-1.14	Stable -	8
Player 4 + Player 8	17.55	3	-0.74	0.17	-0.91	Stable -	5
Player 8 + Player 10	46.82	3	-0.66	0.17	-0.83	Stable -	8
Player 1 + Player 13	26.08	5	-0.50	0.17	-0.66	Stable -	8

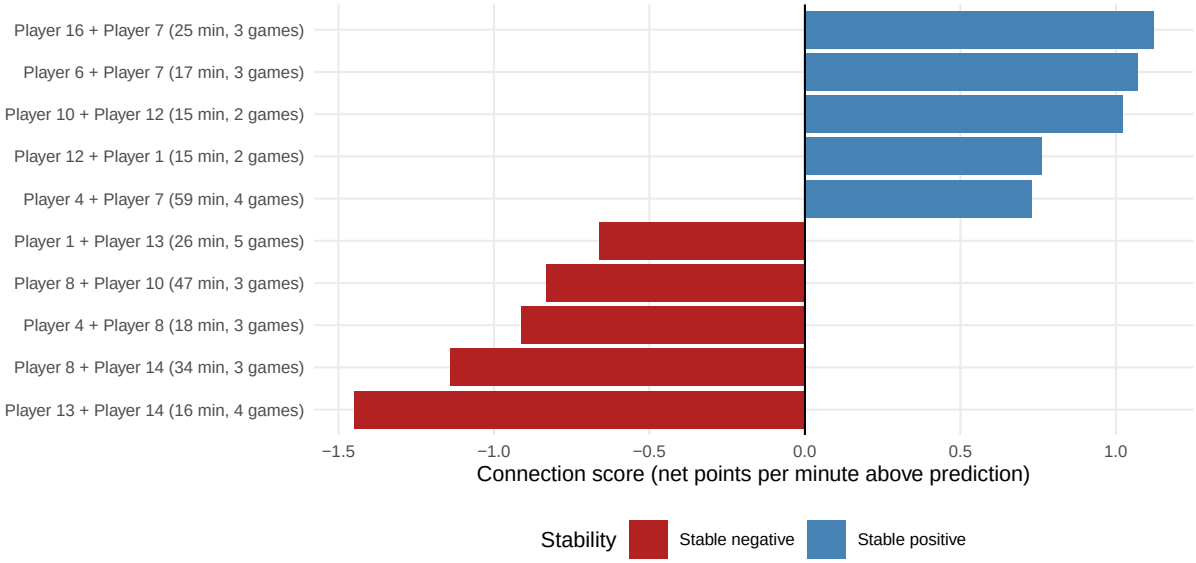


Figure 1: *Connection scores for the five highest and five lowest pairs. Bars are net points per minute above model prediction; colour shows leave-one-game-out stability. Pairs share at least 15 minutes across at least two games.*

Connections within individual games

For Figure 2, the same season-model residuals are averaged within each pair and game. These scores are descriptive and in-sample. They do not use the held-out prediction procedure. A two-minute within-game display threshold is used; this is not a reliability guarantee. Nodes and colours are consistent across panels; a missing edge indicates insufficient displayed exposure, not a score of zero.

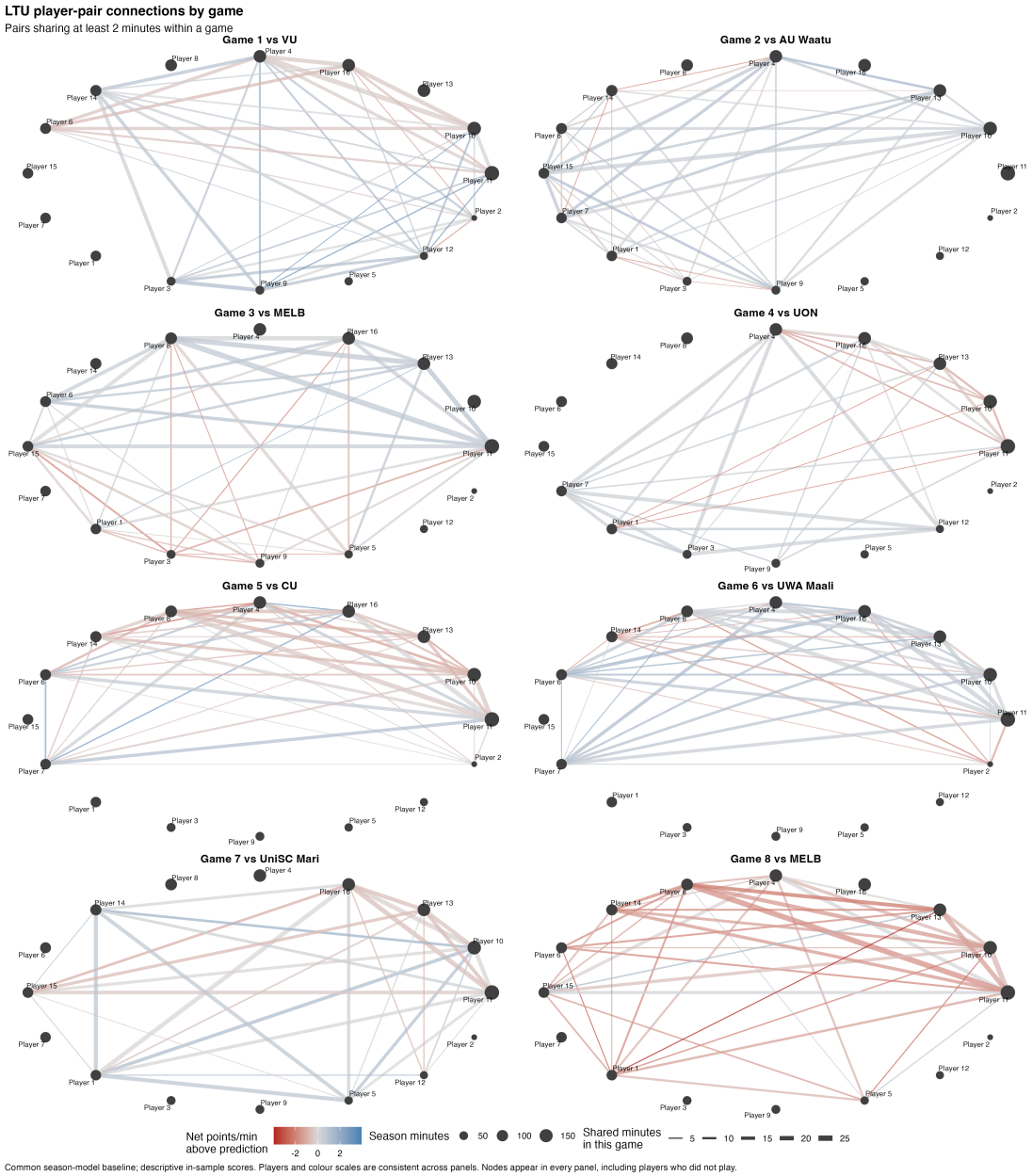


Figure 2: *Pair scores by game using a common season-model baseline. Nodes are repeated in every panel, including players who did not play. Node size is season minutes.*

Connection network and rotation structure

The connection-score network in Figure 3 contains eligible adjusted pair scores. It differs from the usage network used to calculate centrality: the latter uses a threshold of one shared minute in one game, rather than 15 minutes in two games.

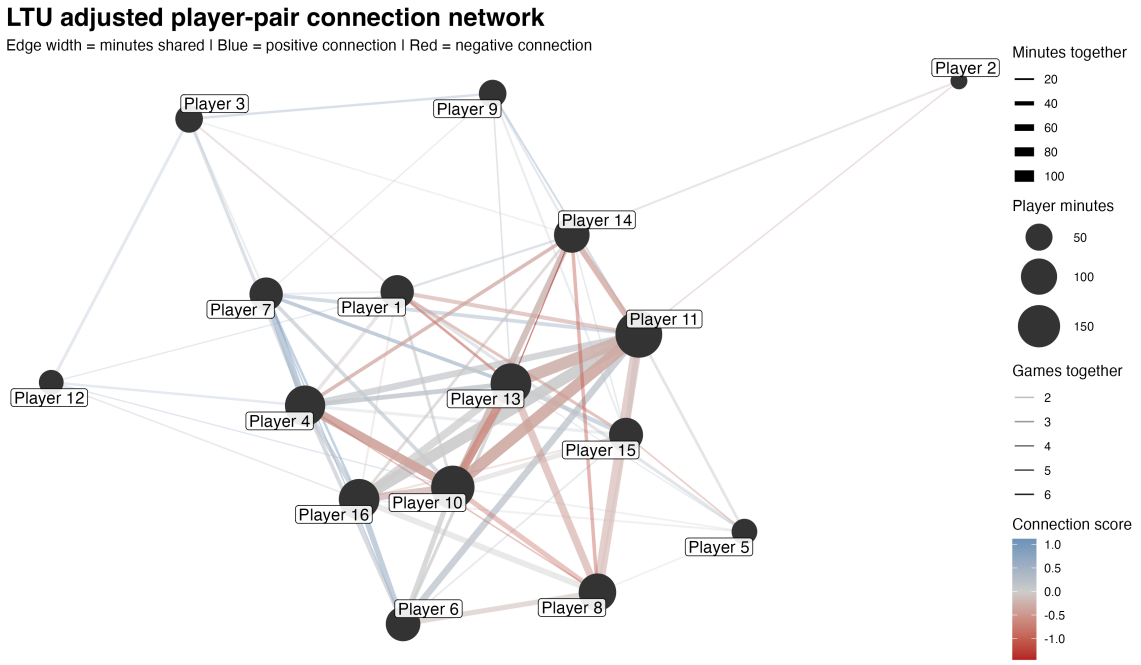


Figure 3: *Network of pairs with a connection score. Nodes are players (size is total minutes), edge width is shared minutes, and edge colour shows the connection score (blue positive, red negative). Players are anonymised.*

In the shared-minute usage network, 112of120 possible pairs shared the court (density 0.933; Table 3). Shared-minute exposure was spread fairly evenly (Gini 0.27), whereas weighted betweenness was highly concentrated (Gini 0.88):11of16 players had a betweenness of zero (Figure 4).

The two highest bridging scores belong to Player 11(0.42) and Player 4(0.19). They rank 1and5 of 16for shared minutes, so the pattern is consistent with heavy exposure. Betweenness is not simply a function of minutes, however:Player 10, second for shared minutes, has a betweenness of 0.00. The pattern describes deployment and does not establish tactical leadership.

Table 3: *Usage-network summary (pairs with at least one shared minute).*

Measure	Value
Players	16

Table 3: *Usage-network summary (pairs with at least one shared minute).*

Measure	Value
Connected pairs	112
Possible pairs	120
Density	0.933
Gini of weighted strength	0.267
Gini of weighted betweenness	0.881
Freeman betweenness centralisation (unweighted)	0.002

Situational coverage

LTU spent 44% of observed minutes leading by at least six points, 44% within five points and 12% trailing by at least six (Figure 5). Trailing situations appeared in only 3 of the retained games. The clutch window contained 14 stints totalling 9.4 minutes across 3 games. Only 3 lineup-by-situation combinations reached five minutes across two games. This is too little to compare closing lineups, so the situational results are descriptive.

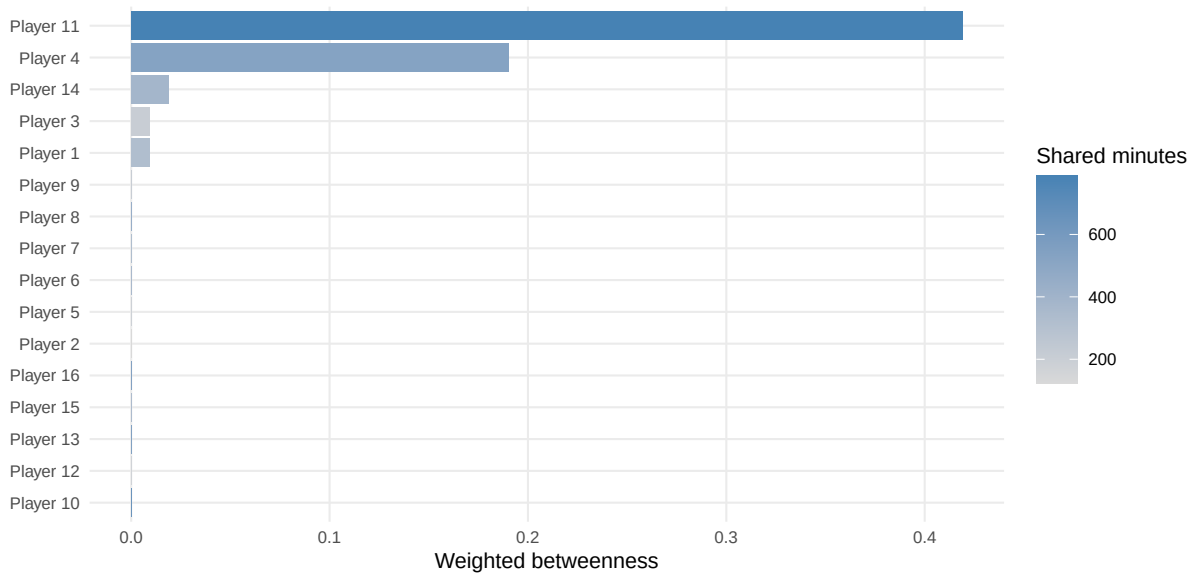


Figure 4: *Weighted betweenness by player in the shared-minute network. Bar shading shows weighted strength (total shared minutes).*

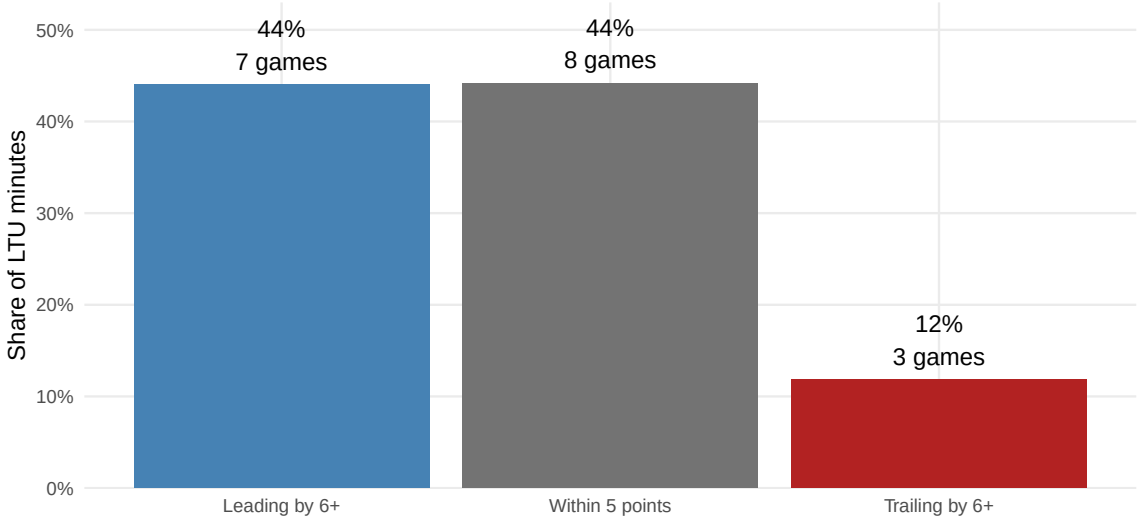


Figure 5: *Share of LTU minutes by score state at stint start, labelled with the number of games in which each state occurred.*

Predictive performance

Across 225 held-out stints, the pair adjustment slightly reduced weighted RMSE from 2.303 to 2.300 (Table 4). The adjustment lowered error in 5 of 8 held-out games (Figure 6). The weighted correlation between the pair feature and the baseline residuals was 0.054. However, the overall improvement was small, and the exploratory permutation test provided no clear evidence of a predictive advantage (p = 0.219). These findings apply to the tested average-pair adjustment and do not establish that all models of player interactions would fail.

Table 4: *Held-out weighted RMSE by game. Difference is baseline minus pair-adjusted error, so positive values mean the pair scores helped.*

Game	Stints	Baseline	With pair scores	Difference	Improved
Game 1 vs VU	29	2.368	2.419	-0.052	No
Game 2 vs AU Waatu	21	2.442	2.440	0.002	Yes
Game 3 vs MELB	32	2.317	2.375	-0.058	No
Game 4 vs UON	32	2.353	2.332	0.021	Yes
Game 5 vs CU	33	2.219	2.159	0.061	Yes
Game 6 vs UWA Maali	30	1.910	1.876	0.034	Yes
Game 7 vs UniSC Mari	23	2.248	2.272	-0.024	No
Game 8 vs MELB	25	2.545	2.496	0.049	Yes
All	225	2.303	2.300	0.003	5 of 8

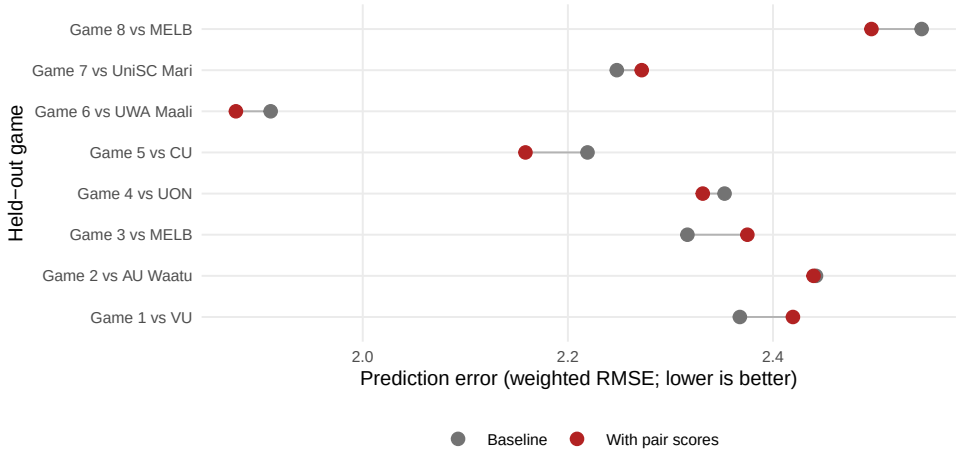


Figure 6: *Held-out prediction error by game. Points left of the grey point (lower error) mean the pair scores helped in that game.*

7 Discussion

The analysis distinguishes descriptive pair performance, rotation structure and predictive usefulness. Above-prediction shared minutes identify combinations that may warrant further investigation, but overlapping lineups prevent attributing those residuals to a particular pair. Leave-one-game-out stability adds evidence about sensitivity to the observed games, while held-out prediction asks whether the information transfers to an omitted game. These are different forms of validation, and only the second looks beyond the games used to estimate the scores.

The lack of predictive improvement limits claims that the pair scores identify repeatable advantages. The strong coefficient shrinkage of the individual ratings reflects the influence of regularisation in this design; it does not mean players have equal ability, nor that the model has isolated their contributions. The network results are descriptive: the rotation was broadly connected, and bridging positions sat with heavily used players.

For coaches, the most defensible use is targeted video review. The pair table can point to stretches in which to examine spacing, defensive coverage and the contribution of the other three players. The usage network can describe which players link rotation groups, without assigning tactical leadership from centrality alone. The results give no basis for changing rotations on predictive grounds.

Limitations

- **Sample:** one season, one team's perspective, and 8 retained games. Excluded games also remove data, and a coverage screen of 50% is permissive.
- **Lineup reconstruction:** opening lineups are inferred, and lineups are carried between periods without independent verification between periods.
- **Outcome measure:** performance is per minute rather than per possession, so pace differences remain, and short stints give noisy rates.
- **Controls:** the opponent control is team-level and regularised, and most opponents appear in only one retained game. Score state is not in the baseline model.
- **Interpretation:** a pair's residual can reflect the contributions of the other three players or game circumstances. Many overlapping pairs create opportunities for extreme rankings by chance, and residuals are in-sample.
- **Validation:** stability labels reuse overlapping games, and the permutation benchmark ignores within-game dependence.

Future work

Future work should prioritise verified lineups and archived source data, then more games and accessible seasons of 2026. Useful sensitivity checks include stricter coverage rules, alternative stint and pair-exposure thresholds, and a comparison of `lambda.1se` with

`lambda.min`. A larger dataset could support possession-adjusted outcomes, controls for opposing lineups and a pair adjustment calibrated within each training fold. A temporal split would better assess prediction of future games than leave-one-game-out within a season.

8 Conclusion

This project develops a workflow for examining player connections and rotation structure in LTU women's basketball. The pair scores identify combinations for further observation, and the usage network describes broad player overlap with concentrated bridging positions. However, the tested pair-score adjustment did not clearly improve held-out prediction.

The practical recommendation is to use these results to prioritise video review and data collection. More verified games, and evidence that patterns repeat, are needed before using the scores to recommend changes to rotations.

Appendix A: Software and reproducibility

Analyses were run in R (R Core Team 2026) using the tidyverse (Wickham et al. 2019), `glmnet` (Friedman, Hastie, and Tibshirani 2010) and `igraph` (Csárdi and Nepusz 2006). The runner, `run-analysis.R`, executes four scripts in order: `R/LTU-team.R` loads saved inputs, reconstructs stints and calculates connection scores and situational summaries; `R/LTU-prediction.R` evaluates held-out predictions; `R/LTU-network-by-game.R` produces per-game networks; and `R/LTU-league-comparison.R` calculates league-level centrality scores.

The report is rendered separately and reads saved outputs and input snapshots without downloading new data. Player labels are used in all tables and figures displayed in the report.

Appendix B: League context

For context, a team-level network was built from all UBL results in 2025 competition. Each match adds an arrow from the losing team to the winning team, weighted by winning margin, and eigenvector centrality ranks teams so that wins over highly ranked teams count for more (Keener 1993; Bonacich 1972). Scores are scaled so the top team equals 1. The Gini coefficient summarises how concentrated these scores are.

In the women's competition UTAS ranked first and LTU ranked 3 of 12 (Gini 0.53). In the men's competition UON ranked first (Gini 0.57) and UTAS ranked 4, so team strength was specific to each competition (Figure 7). This is one season, and the measure rewards winning margins as well as wins.

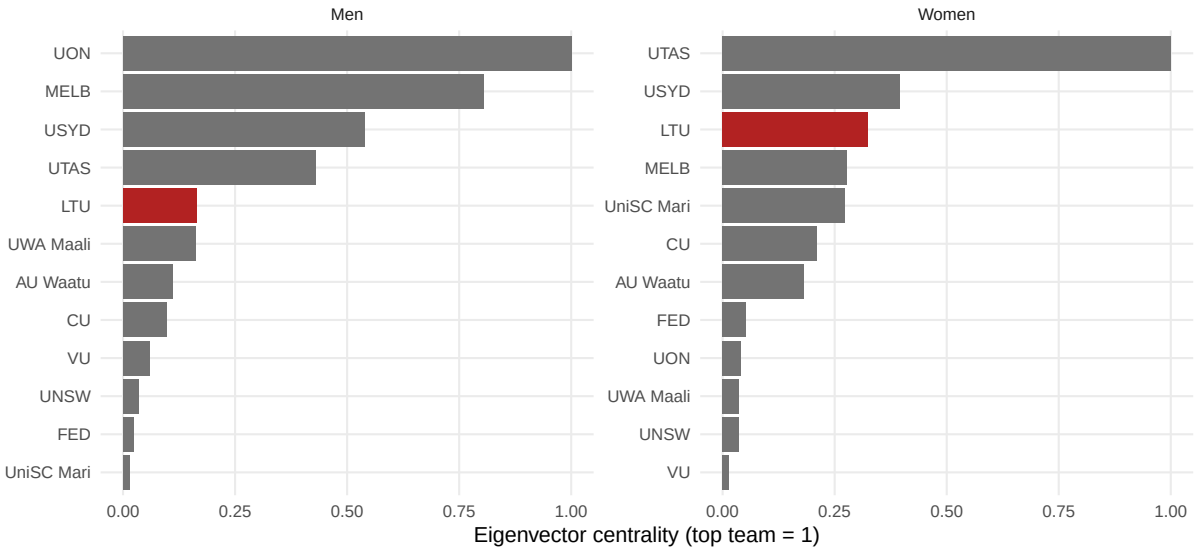


Figure 7: *Eigenvector centrality of teams from the 2025 results network, by competition. LTU is highlighted.*

References

Bonacich, Phillip. 1972. “Factoring and Weighting Approaches to Status Scores and Clique Identification.” *Journal of Mathematical Sociology* 2 (1): 113–20. https://doi.org/10.1080/0022250X.1972.9989806.

Csárdi, Gábor, and Tamás Nepusz. 2006. “The igraph Software Package for Complex Network Research.” *InterJournal* Complex Systems: 1695. https://igraph.org.

Freeman, Linton C. 1978. “Centrality in Social Networks: Conceptual Clarification.” *Social Networks* 1 (3): 215–39. https://doi.org/10.1016/0378-8733(78)90021-7.

Friedman, Jerome, Trevor Hastie, and Robert Tibshirani. 2010. “Regularization Paths for Generalized Linear Models via Coordinate Descent.” *Journal of Statistical Software* 33 (1): 1–22. https://doi.org/10.18637/jss.v033.i01.

Hoerl, Arthur E., and Robert W. Kennard. 1970. “Ridge Regression: Biased Estimation for Nonorthogonal Problems.” *Technometrics* 12 (1): 55–67. https://doi.org/10.1080/00401706.1970.10488634.

Keener, James P. 1993. “The Perron–Frobenius Theorem and the Ranking of Football Teams.” *SIAM Review* 35 (1): 80–93. https://doi.org/10.1137/1035004.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://doi.org/10.32614/R.manuals.

Rosenbaum, Dan T. 2004. “Measuring How NBA Players Help Their Teams Win.” 82games.com. https://www.82games.com/comm30.htm.

Sill, Joseph. 2010. “Improved NBA Adjusted +/- Using Regularization and Out-of-Sample Testing.” In *MIT Sloan Sports Analytics Conference*. https://cdn.prod.website-files.com/68d6be744d7efccc2207f571/68d6be744d7efccc220807d7_SSAC10%20-%20Basketball_Sill.pdf.

Wasserman, Stanley, and Katherine Faust. 1994. *Social Network Analysis: Methods and Applications*. Cambridge University Press. https://doi.org/10.1017/CBO9780511815478.

Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.

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